Semester Project: AI-Powered Social Media Content Moderator

Assignment Group B: Binary Classification with XLM-RoBERTa

Course: Artificial Intelligence (CS-302) Fall 2025

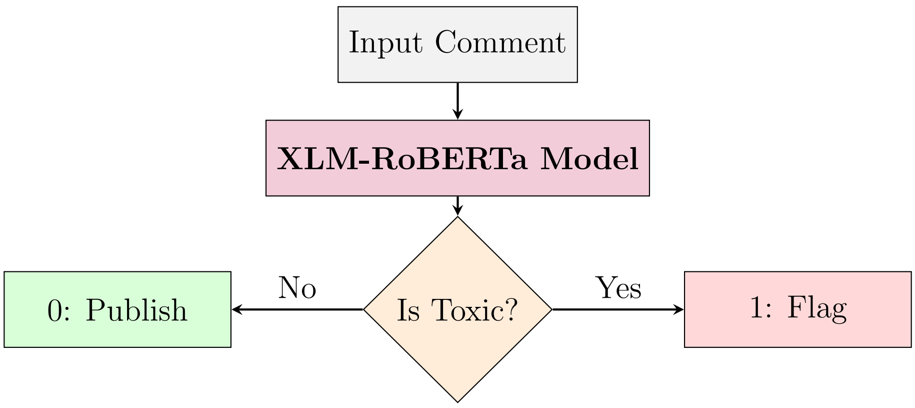
1 Project Overview

Your group is tasked with building the "Digital Gatekeeper" for a social media moderation system. Social media platforms in Pakistan are flooded with code-mixed comments (English + Urdu + Roman Urdu). Manual moderation is impossible at this scale.

You must build a Deep Learning model that analyzes incoming comments and automatically flags them as **Safe (0)** or **Flagged (1)**.

Goal: A high-recall system that catches as much toxic content as possible without blocking innocent conversations.

2 System Architecture



3 Dataset Specifications

You will use the RUHSOLD dataset (Roman Urdu Hate-Speech and Offensive Language Detection).

- **Source:** GitHub Repository
- **Download Link:** `https://github.com/haroonshakeel/roman_urdu_hate_speech`
- **Files to Use:**
  1. `task_1_train.tsv` (Use for Training)
  2. `task_1_test.tsv` (Use for Final Evaluation)
- **Format Note:** These are **.tsv** (Tab Separated Values) files. Load them in Pandas using:

```
pd.read_csv('task_1_train.tsv', sep='\t')
```

- **Label Mapping:**
  - **Normal** → Class **0** (Safe)
  - **Hostile / Offensive** → Class **1** (Toxic)

4 Technical Requirements (Strict)

To ensure consistency across the class, you must adhere to the following stack:

4.1 Model Architecture

- **Assigned Model:** `xlm-roberta-base` (XLM-R)
- **Library:** Hugging Face `transformers`
- **Why?** XLM-RoBERTa is a newer, more advanced model than BERT. It is specifically optimized for low-resource languages and handles spelling variations (like the non-standard spellings in Roman Urdu) significantly better than older models.

4.2 Preprocessing Pipeline

Before feeding text to the model, apply these cleaning steps:

1. **Lowercase:** Convert all text to lowercase to standardize Roman Urdu (e.g., "Main" → "main").
2. **Clean Noise:** Remove usernames (@user), URLs (http://...), and emojis.
3. **Tokenization:** Use the standard XLM-RoBERTa Tokenizer (SentencePiece).
4. **(Bonus) Lexicon Feature:** The repo contains `Hate_Words_RomanUrdu_Lower.txt`. You may use this list to add a "Contains Hate Word" flag to your input.

5 Deliverables

Your submission must include the following in a ZIP file:

5.1 Jupyter Notebook (.ipynb)

Must be clean, commented, and runnable. It should contain:

- Data Loading Visualization (Show class distribution).
- Preprocessing Functions.
- Model Training Loop (using PyTorch or Trainer API).
- Evaluation on the Test Set.

5.2 Performance Report (PDF)

1. **Classification Report:** Precision, Recall, and F1-Score for Class 0 and Class 1.
2. **Confusion Matrix:** A visual heatmap showing True Positives vs. False Positives.
3. **Failure Analysis:** Identify 3-5 examples where your model failed (e.g., a Safe comment marked as Toxic) and explain *why* you think it happened.

5.3 Saved Model

Save your fine-tuned model weights (e.g., `model.pt` or `/saved_model` folder) and upload them to Google Drive/One Drive. Include the link in your report.

6 Reference Material

For a deeper understanding of the dataset creation and the challenges of Roman Urdu, refer to the documentation paper:

> *Rizwan, H., Shakeel, M. H., & Karim, A. (2020). Hate-Speech and Offensive Language Detection in Roman Urdu. Proceedings of EMNLP 2020.*